

From Thoughts to Interaction: Designing Controls for Video Playback Gestures with Embodied Schemata

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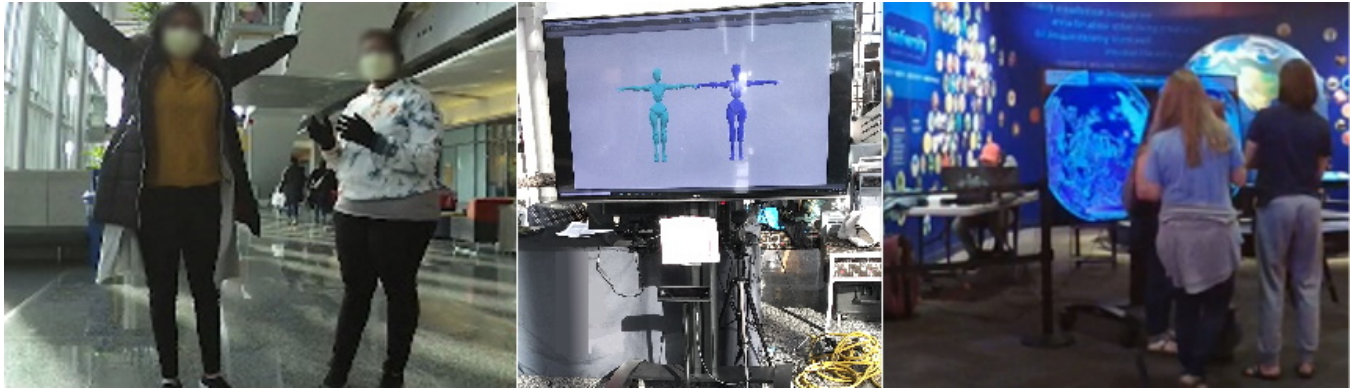


Figure 1: People interact with public displays in a variety of settings. From left to right: users jumping to control the screen at a student center; playing with their avatars near a coffee shop; and, museum visitors exploring a globe-based data visualization using gestures.

ABSTRACT

Interactive public displays are now pervasive; however, designing gestures to interact with them is still a challenge for embodied interaction. We introduce a methodology, based on Conceptual Metaphor Theory (CMT) and Elicitation Studies, that can inform the work of interaction designers when they craft gestures that "make sense" to their users. Our approach is structured in three phases: elicitation, analysis, and design. Through the use case of an interactive video player, we describe how we collected data from participants in an elicitation study, how we extracted the embodied schemata that users associated with each video function, and how an interaction designer can use those schemata to design multiple gestures. We then present the results of a survey that show that the gestures generated with embodied schemata made more sense to people than those crafted without using embodied schemata.

CCS CONCEPTS

• **Human-centered computing** → HCI design and evaluation methods;

KEYWORDS

Design Methods, Embodied Interaction, Gestural Interaction

ACM Reference Format:

Adina Friedman and Francesco Cafaro. 2023. From Thoughts to Interaction: Designing Controls for Video Playback Gestures with Embodied Schemata. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3585769>

1 INTRODUCTION

From touchscreens [16] to motion-sensing devices [44], we can interact with public displays far beyond the traditional windows-icon-mouse-pointer (WIMP) paradigm –see Figure 1. These novel, embodied interactions [15, 20] offer exciting opportunities for presenting enjoyable and memorable experiences to casual users [8]. Although interactive displays are now pervasive in public spaces like museums, airports, and public squares [9, 11, 17, 44], designing gestures to control them is still a challenge for embodied interaction [12, 30].

Arguably, in-lab elicitation studies [50] are the state-of-the-art for gesture design. During elicitation studies, potential users are exposed to animations that represent functionalities of the system, and asked to recommend how they would control that functionality. Even gesture sets generated with elicitation studies, however, may be problematic. First, gestures may be disconnected from each other [46]; this can be disorienting for the users [9]. Second, some interactive displays may require multiple gestures to activate the same functionalities [36] –for example, to maximize the chances that

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CHI EA '23, April 23–28, 2023, Hamburg, Germany
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ACM ISBN 978-1-4503-9422-2/23/04.
<https://doi.org/10.1145/3544549.3585769>

visitors will notice that a screen is interactive in crowded museum galleries [35]. In contrast, elicitation studies typically match one winning gesture to each function and cannot be used to inform the design of multiple, alternative gestures.

In this paper, we introduce a new methodological approach to gesture design that can inform the work of interaction designers. The theoretical underpinning of this work comes from cognitive linguistics, in particular Lakoff and Johnson's Conceptual Metaphor Theory (CMT) [31, 32]. According to CMT, we acquire basic mental patterns ("embodied schemata") at a very early stage of our life through our bodily interaction with the world [27]. For example, a baby learns the BALANCE schema when they are eventually able to stand upright [27]. Embodied schemata can be particularly useful for designing embodied interactions because they are basic mental patterns that are common across many languages and cultures [32] (while more complex patterns, like conceptual metaphors, are culturally-dependent [26]). Thus, their simplicity provides a framework for designing the interaction in public places that attract a diverse crowd of visitors.

To illustrate how embodied schemata can be used to facilitate the rapid design of gestures, we structured our paper on the premise that "the best way to talk about methods is to show instances of the actual work" [28]. We chose an interactive video player as our case study, to facilitate comparisons with earlier works on elicitation studies that focused on controlling interactive media and smart TVs (e.g., [14, 36, 46]). Indeed, many public displays use videos to convey information [18], having them play on a continuous loop. Without a means to control them, if a user does not show up at the right time, they have to wait for the video to restart to see what they may have missed, and the only way to repeat a segment is by waiting for the video to loop back around again. A solution to this would be to develop intuitive tools [49] that allow users to change the video playback as needed.

The methodological approach that we introduce in this paper is structured in three phases: elicitation, analysis, and design. We start with (1) a traditional elicitation study [50] accompanied by, (2) an analysis of participants' recommendations to identify what schemata [32] people were using for each function. These schemata are then used to (3) design multiple gestures. Finally, we conducted a crowd-sourced survey to evaluate whether gestures based on the schemata gathered from the elicitation study "made more sense" to people than other gestures crafted without the schema. We present the results of a study with 39 total participants, 10 in the initial elicitation study and 29 in the evaluation survey. Our results indicate that embodied schemata enable interaction designers to craft gestures that make more sense to their users.

2 RELATED WORK

2.1 Embodied Interaction

Our work is on embodied interaction [15]. According to Dourish, we construct meaning through our embodied (i.e. physical, and situated in time and space) interaction with the world [15]. Hornecker (e.g., [20]) highlights the role of the user's body, and argues that "movement and perception are tightly coupled." In this paper, we use the words "embodied interaction" to denote interactive displays

and installations that are controlled by hand gestures and body movements (similarly to the definitions in [11, 20]).

2.2 Interacting with Public Displays

Advances in display technology have led to the availability of large displays: they can facilitate collaboration and analysis of information [29, 38, 40], deliver public information to citizens [13], promote data exploration in museums [12], create shopping opportunities at airports [4], etc.

As these new technologies emerge, it becomes necessary to develop new interaction methods, especially when the traditional mouse and keyboard controls may prove impractical [38, 39]. Fortunately, with smaller sensors such as the Xbox Kinect and ultrasonic sensors, gesture-based control systems have become more viable for commercial (and personal) interactive systems [1, 11, 17, 29]. These devices are small and can be attained at a low enough price point that it opens up gesture-based interaction to a larger audience than ever before.

Previous work designing gesture interfaces has tried to find gestures that are simple and appeal to most users [29, 39]. There has also been some examination into how to lure users towards a gesture-based display and how their interactions are affected by their representation on screen [44]. Our work contributes to this line of research with a methodological approach that enables interaction designers to craft gestures rapidly.

2.3 Elicitation Studies

Our work is based on elicitation studies. Originally defined by Wobbrock in [49], "Guessability" (or "Elicitation") studies have end-users (not expert designers) design interactive systems. Guessability Studies have been applied to a variety of design domains. For instance, in [42], a Guessability study led to the creation of a set of motion gestures for smartphones; Piumsomboon et al. reported the result of a Guessability study that focused on gestures for augmented reality [41]; and Morris et al. used a Microsoft Kinect to control a web browser on a TV screen without a remote [36]. More recent work has examined the use of online surveys for system design in place of lab-based elicitation studies [3]. Research has also focused on improving the discoverability [10] or memorability [2] of gestures generated with elicitation studies. A 2020 literature review found over 210 papers on elicitation studies [47].

These works, however, have not directly examined the relationship between gestures and schemata [29, 39]. Additionally, they do not typically include designers in the process. Our methodology builds upon elicitation studies by introducing an approach that re-incorporates the work of interaction designers into the elicitation process.

2.4 Using Conceptual Metaphor Theory (CMT) to Design Interactive Systems

Our methodological approach is rooted in Lakoff's and Johnson's Conceptual Metaphor Theory (CMT) [31, 32]. CMT is grounded in embodied cognition [48] a theory that posits that our body plays a fundamental role in our cognitive processes.

CMT has been used to inform the design of "intuitive" [5] embodied interactions. Embodied schemata have been used to map

input actions into perceptual feedback [6]. The work in [33] explored pairings between physical attributes of objects (e.g., a big vs. small cube) and metaphors (e.g., powerful vs. weak). Although these works greatly inform our methodology, they typically require 1:1 direct matching between gestures and functionality (e.g., "fast" body movements to "fast" tempo in a music installation [6]).

In closely related work, CMT was used to inform the design of tangible user interfaces at a beverage company: the work in [23] describes a low-fidelity prototype of a small plastic ball that can be manipulated to open and post invoices. Hurtienne [21] discusses strategies to "source" embodied schemata that are then used in the design process. Work in [7] describes a design process to "instantiate" individual embodied schemata into physical objects. Designers can then use those objects to make sense of what each specific schema entails. The investigation in [25] explores how designing with embodied schemata may produce gestures that are accessible by users of different age ranges. These methodological explorations inform the work that we discuss in this paper, but they entail relatively long study and analysis sessions that may be impractical for some designers and practitioners. In this late-breaking work paper, we propose a method that can be used to rapidly inform the design of multiple gestures to interact with public displays.

3 METHODOLOGY

3.1 Step 1 - Eliciting Gestures from Participants

The first step of our design methodology is very similar to a typical elicitation study, as defined in [50].

In our case study (media player on a public display), the elicitation study took the form of remote interviews conducted by the researcher. The first portion of the study consisted of 10 participants, 8 female and 2 male ranging in age from 20 to 70 years old, three of which were non-native English speakers. Further personal information was not collected.

Subjects were asked to demonstrate gestures they would use for four pairs of video controls (eight functions in total): Play video/Pause video; Fast forward/Rewind; Zoom in/Zoom out (on the video); Volume louder/Volume softer.

Differently from typical elicitation studies [50], participants were asked to think aloud and explain why they thought the gestures they selected were appropriate for the task of controlling the video feed.

Participants were recruited through social networks and interviewed over Zoom. The interviews lasted less than 10 minutes, with the overall purpose of the interview being explained afterward.

3.2 Step 2 - Identifying Schemata

Elicitation studies use agreement measures to assign the gesture recommended by most participants to a functionality of the system [47]. Rather, our analysis focuses on identifying recurrent embodied schemata that designers can use to rapidly generate multiple gestures.

3.2.1 Analysis Methods. A researcher experienced with embodied schemata analyzed the video recordings and the transcripts of participants' remarks, with the aim of coding which schema participants were using when recommending their gestures. In

line with the literature on designing with embodied schemata (e.g., [1, 24]), we used the catalog created by Hurtienne in [22] as our coding dictionary. A common example was participants swiping their hands vertically to control the volume, which we associated with the Up-Down schema, with 'up' being used to represent louder and 'down' being used to represent softer. For pause, participants held up a hand palm out in a 'stop' gesture, which we associated with the Blockage schema. To activate the play function, participants indicated that they wanted to use some sort of gesture to prompt the video to continue. We associated this intention with the Compulsion or Enablement schema.

The coding was done after the interviews; however, during the interviews, there were times when the moderator asked clarifying questions to make sure she understood the subject's thought process and the reason why they thought a gesture was appropriate. We analyzed around 40 elicited gestures, and identified 8 distinct schema.

In order to rapidly produce gestures designs, we did not use multiple researchers to individually code the transcripts. Rather, using an approach similar to Interaction Analysis [28], another researcher (experienced with embodied schemata) discussed extracts from the interview video and transcripts with the primary researcher. Instances of disagreement were resolved during two one-hour meetings.

3.2.2 Results: Schemata Used for Each Function. Table 1 lists all the schemata on which participants grounded their recommendations during the elicitation study.

For the Play function, five participants used the Compulsion schema (recommending motions that indicated propelling the content further), and 3 used the Enablement or Restraint Rem oval schema (their replies could be categorized into either one). The use of the Compulsion schema relies on the idea of telling the video to go on or to play. The participants who used the Enablement/Restraint Removal schemata described pausing the video as introducing a blockage and allowing it to play again as removing that blockage. For instance, one participant phrased it as "showing motion rather than having a barrier". The Pause function yielded some of the most consistent results; 8 out of 10 participants said that they would pause the video by holding their hand out flat with their palm towards the screen. This is a gesture that is commonly used in conversations in the U.S. as a way to say stop, so many participants carried that association. Their comments indicated usage of the Blockage schema. Similarly, 8 out of 10 participants used the Left-Right schema for the fast-forward and rewind functions. Most indicated that they associated left with going back in the video and right with going forward in the video. One participant compared it to "flipping forward and back in a book." For the zoom in and zoom out functions, 7 participants used the Big-Small schema. This was based on the idea of expansion and contraction or stretching and compressing an object. The participants who used these ideas chose gestures such as pulling their hands apart or spreading their fingers to enlarge an image, and pushing their hands back together or pinching their fingers to zoom out. Two participants used the Near-Far schema, using the idea that being closer allowed them to see images at a larger scale while being farther made the images appear smaller. The volume functions were matched to an Up-Down

Function	Play	Pause	Fast Forward	Rewind
Schema: number of participants who used it	Compulsion: 5 Enablement: 3 No Schema: 2	Blockage: 8 No Schema: 2	Left-Right: 8 Front-Back: 2	Left-Right: 8 Front-Back: 2

Function	Zoom In	Zoom Out	Volume Louder	Volume Softer
Schema: number of participants who used it	Big-Small: 7 Near-Far: 2 Up-Down: 1	Big-Small: 7 Near-Far: 2 Up-Down: 1	Up-Down: 5 Left-Right: 4 No Schema: 1	Up-Down: 5 Left-Right: 4 No Schema: 1

Table 1: System Functionalities and the Schemata Participants Used during the Elicitation Study

schema by 5 participants. They based this on common terms such as “turning the volume up”. Four participants used a Left-Right schema for volume control, specifically looking at it as clockwise and counter-clockwise rotations. Two participants compared this to using knobs to adjust volume “like a car radio” or “an old TV.”

Table 2 provides an overview of the recommended matching between schema and function pairs. These recommendations include a single schema for each function pair (the exception is the Play/Pause function which uses two schemata). This is the final output of the elicitation study.

3.3 Step 3 - Using Schemata to Design Gestures

In the next step, the researchers (who have years of experience in interaction design) created 4 gestures for each functionality of our media player (play/pause, fast forward/rewind, etc). We produced a total of 16 gestures (four for each function pair). Half of the gestures was based on schemata matching while the other half was not created to match any specific schema. All gestures were designed to be performed with the user standing in one spot in front of a 65" TV, mainly using their arms and upper body.

The gestures designed are listed below. Gestures marked with * are grounded on the schema for each function.

- **Volume Louder/Volume Softer (Up-down):** Swipe up/down*, Pointing up/down*, Swipe left to right/right to left, Turning left/right
- **Zoom In/Zoom Out (Big-Small):** Spreading hands/Bringing hands together*, Spreading fingers/Pinching fingers*, Clapping, Leaning left/right
- **Fast Forward/Rewind (Left-Right):** Swipe left to right/right to left*, Reach right/reach left*, Waving hands above head/below head, Arms forward/Arms back
- **Play/Pause (Compulsion/Blockage):** Wave forward/Palms out*, Arms forward/Cross X*, Arms over head left/right, Tip-toes/Bend knees

Function Pair	Recommended Schema
Play/Pause	Compulsion/Blockage
Fast Forward/Rewind	Left-Right
Zoom In/Zoom Out	Big-Small
Volume Louder/Volume Softer	Up-Down

Table 2: Schema recommendations for function pairs

4 EVALUATING THE DESIGNED GESTURES

Our hypothesis was that the gestures based on the embodied schema would “make more sense” to people. To verify our hypothesis, we compared the two sets of gestures through an online survey (those created with and without the aid of the schemata in Table 2).

4.1 Survey Methodology

To compare gestures based on the embodied schemata vs. other gestures designed by the researchers, we conducted a survey using Amazon’s Mechanical Turk. Thirty participants were recruited using Amazon Mechanical Turk. The survey gave us 29 legitimate results, with one incomplete survey eliminated. We used a within subject design, so all 29 participants were exposed to the same set of gestures. For each function pair, participants watched a video with 4 gestures (an example is shown in Figure 2), 2 of which were based on the schema matching in Table 2, and 2 that were not. Participants were then asked to rank the gestures according to “how well they made sense (to you) to use for [function].” The phrase “make sense to you” was used to indicate that participants did not need to directly understand the researchers’ reasons for using certain gestures (based on embodied schemata). Participants only needed to have their own (non-specific) reason for judging whether gestures were appropriate for system functions.

4.2 Survey Results

For each of the four function pairs in Table 2, we analyzed the rankings assigned by participants to each of the four gestures they were exposed to. As we described above, two of those gestures were designed using the embodied schemata that we identified from our elicitation study, while two were not. Because each participant was asked to rate each of the four gestures (i.e., within-subject design), we used a Friedman Test for our analysis (basically, the non-parametric version of an ANOVA with repeated measures). We considered whether the gesture was based on embodied schemata or not as our independent variable, with the ranking as our dependent variable. A Friedman test revealed a statistically significant difference in the users’ ranking of gestures as making sense to them for the Fast Forward/Rewind functionality, $\chi^2(2) = 8.28, p = 0.041$. Post hoc analysis with Wilcoxon signed-rank tests was conducted with a Bonferroni correction applied, resulting in a significance level set at $p < 0.0083$ (i.e., 0.05 divided by 6 comparisons). There was a statistically significant difference between the rankings assigned to one of the gestures designed with embodied schemata (swipe left to right) and another created by the researchers not to follow the

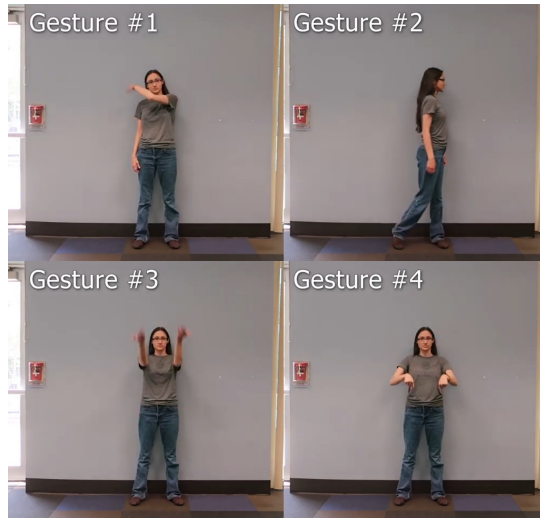


Figure 2: Still from the video for volume control showing the 4 gesture options.

schemata (moving arms forward/backward), $Z = -2.640, p = 0.008$. Additionally, the Friedman test revealed a statistically significant difference in the users' rankings depending on the gesture for the Play/Pause functionality, $\chi^2(2) = 14.40, p = 0.002$. Post hoc analysis with Wilcoxon signed-rank tests was conducted with a Bonferroni correction applied, resulting in a significance level set at $p < 0.0083$. There was a statistically significant difference between the rankings assigned to the two gestures designed with embodied schemata (wave forward/palm out and arms forward/Cross X), and one created by the researchers that did not align with the schemata (Tiptoes/Bend knees), $Z = -1.897, p = 0.008$ and $Z = -2.805, p = 0.005$, respectively. In other words, in these cases, the gestures designed with embodied schemata obtained statistically significantly higher rankings than the others. These results indicate that our methodological approach aided the researchers in crafting gestures that make sense to their users.

5 DISCUSSION

5.1 Matching Schemata to Functions

All of the video functions were in pairs where each operation reversed the other (e.g., fast forward/rewind). Interestingly, during the elicitation study, this led to the same schema being used for both functions in 3 out of 4 pairs (e.g., left/right for fast forward/rewind). This shows that potential users want to use matching gestures to represent corresponding functions. Thus, the task of crafting matching gestures for corresponding functions should be left to designers, rather than to typical elicitation studies that may produce disconnected gesture sets, as shown in [46]. To create gestures that "make sense" to people, designers should carefully craft gestures for corresponding functionalities (like fast forward/rewind) to incorporate the same overarching schema (e.g., left/right).

5.2 Designing Gestures that are Difficult to Elicit

Our methodology may be particularly well suited in situations where participants in elicitation studies cannot easily converge on a gesture. In our use case, the function that participants struggled with the most was the Play function. Many participants took longer to come up with a gesture they felt was appropriate to play the video than the other functions. During the initial interviews, participants often used words that indicated compulsion but struggled to find gestures that they thought were appropriate. When encouraged to talk through their thoughts, participants gestured as they spoke and referenced gestures they would use in conversation with other people. This suggests that users may unconsciously know what schema they want to use but may not be able to come up with gestures for it on their own. In a traditional elicitation study, this could result in a lack of a single cohesive gesture preference for a function (resulting in low levels of the "agreement" metric [50]). On the contrary, when using our approach, the analysis phase (Step 2) enables designers to still learn from users how they want to interact with a system and define candidate gestures matching the users' intention.

5.3 Dealing with Outliers: Gestures that Do Not Fit a Schema

During our elicitation study, two participants suggested gestures for functions that did not fit into a schema defined by Hurtienne [22]. Two participants said they would want to pause and play the video by performing a gesture repeatedly. Another participant said her gesture for volume louder would be to gesture to her ear, and to turn the volume softer she would put her hands over her ears. She said these were the gestures she would use in conversation. In other words, these two participants made recommendations that may be better analyzed through the lenses of McNeill's taxonomy of gestures that people perform during speech [34], rather than with Conceptual Metaphor Theory (CMT). In particular, these two examples could be seen as instances of Metaphoric (they represent an abstract concept) and Beats (they mark specific elements of a conversation) gestures [34]. These instances were limited in our application scenario; future studies should assess if other application scenarios require more complex theoretical structures (e.g., conceptual metaphors [32]) than the simpler (i.e., related with basic bodily experiences) embodied schemata.

5.4 Legacy Bias

Legacy bias is a common problem in gesture elicitation studies: users tend to recommend gestures that they are already familiar with [19, 37]. Regarding our specific scenario, we need to acknowledge that video control conventions are pervasive. As a result, the gestures that people recommended might come from a blend of people's previous experience with video controls, and embodied schemata. In other application scenarios (e.g., when participants are exposed to novel interactive systems they are not familiar with), however, legacy biases can derail the elicitation study. For example, the work in [9] reports that participants to an in-lab elicitation study recommended using a pinch-and-zoom gesture to interact with a data visualization, but that gesture was not discovered at all by the actual users when the gesture-based installation was

deployed in a museum. Thus, future studies should explore how legacy bias may impact our approach.

5.5 Limitations

We want to acknowledge limitations that researchers, designers, and practitioners should consider when adopting the approach in this late-breaking work. We collected embodied schemata from a pool of 10 participants. Although we felt we had reached theoretical saturation [43] with our participants, future studies should assess if the specific schemata that we identified are consistent across different populations and with larger pools of participants. Additionally, embodied schemata are basic mental patterns that are common across many languages and cultures [32], but how people use them to suggest interactions with public displays may be culturally dependent. For example, the preference for using swiping motions from left to right/right to left for the Fast Forward/Rewind function might be related to the fact that the surveys were conducted in English. In English, there is a common association between left-right and past-future [45]. Interestingly, our users preferred a gesture based on left-right schema over one based on front-back, even though phrases like "go back in time" are also common in English. Finally, our elicitation study was conducted remotely, over Zoom (rather than in-lab). Although this approach allowed for more rapid study sessions, future studies should investigate whether an elicitation study with participants standing in front of a 65" display might yield different results. Similarly, we conducted our evaluation using a crowd-sourced platform, so we could not observe if participants performed our gestures before ranking them. We would have asked them to do so in a lab, as it could give participants an improved sense of which gestures feel more "natural" (which may influence our metric – "making sense").

6 CONCLUSION

In this paper, we introduced a methodological approach, based on elicitation studies and Conceptual Metaphor Theory (CMT), that can aid interaction designers in their work of crafting gestures for embodied interaction. Our results showed that our methodology helped in crafting gestures that make sense to people: some of the gestures generated with embodied schemata outperformed those designed not to follow the schemata. Future work should explore the use of this method with larger and more diverse pools of users, and for other types of interfaces.

ACKNOWLEDGMENTS

The researchers would like to thank the participants, and Ruth Siegel for helping film the gesture demonstration videos.

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